

Frontend: Theory Assignment 2

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1. What is CSS Box Model?

The CSS box model tells us how an HTML element takes up space on a webpage. It has four parts: content, padding, border and margin.

2. What is media query? Explain with a sample code.

A media query is used to apply different CSS styles depending on the screen size. It is useful for making a website responsive.

```
@media (max-width: 600px) {  
  body {  
    font-size: 14px;  
  }  
}
```

3. What are the features of Bootstrap?

Bootstrap provides ready-made CSS classes and components. It includes a responsive grid, buttons, forms, cards, navbar, utilities and other UI components.

4. Write an example of basic grid structure in Bootstrap.

A basic Bootstrap grid contains a container, a row and one or more columns.

```
<div class="container">  
  <div class="row">  
    <div class="col-md-6">Column 1</div>  
    <div class="col-md-6">Column 2</div>  
  </div>  
</div>
```

5. What is CSS Selectors?

CSS selectors are used to select HTML elements that we want to style. Examples are element selector, class selector, ID selector, universal selector and attribute selector.

6. What are CSS backgrounds, list the properties?

CSS background properties are used to set the background of an element. Some common properties are background-color, background-image, background-repeat, background-position, background-size and background-attachment.

7. How can we overlap elements in CSS?

We can overlap elements by using position properties such as relative and absolute. The z-index property can be used to decide which element appears on top.

8. What are the different positioning properties in CSS?

The main CSS position properties are static, relative, absolute, fixed and sticky.

9. What is ECMAScript?

ECMAScript is the standard specification for scripting languages like JavaScript. JavaScript follows the ECMAScript standard.

10. What is the difference between var, let and const in JavaScript. Explain Temporal Dead Zone?

var: It is function-scoped and can be redeclared and reassigned.

let: It is block-scoped and can be reassigned, but cannot be redeclared in the same scope.

const: It is block-scoped and cannot be reassigned. It must be given a value when declared.

Temporal Dead Zone: It is the time between entering a block and declaring a let or const variable. During this time, accessing the variable gives a ReferenceError.